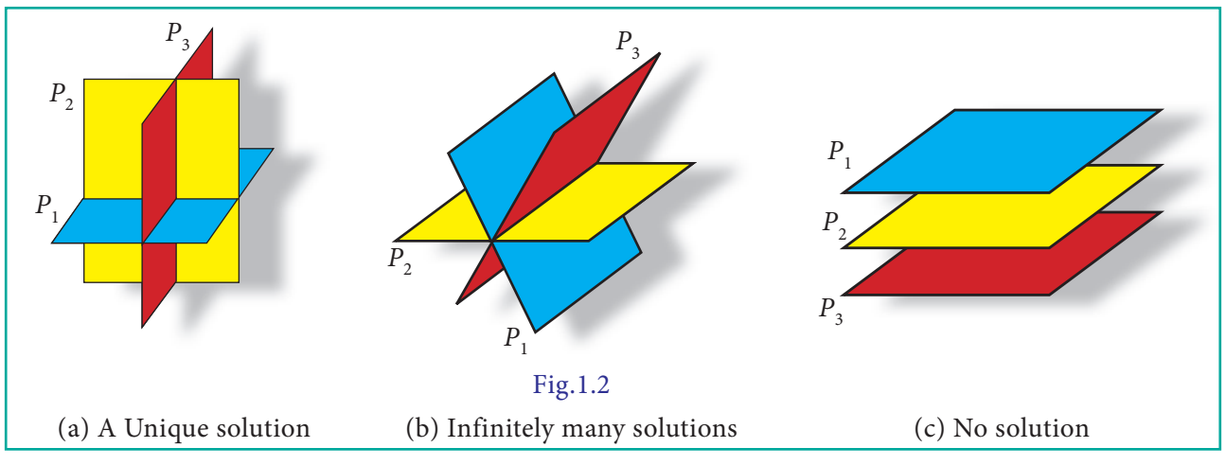


Fig.1.2

(a) A Unique solution (b) Infinitely many solutions (c) No solution

Note

Every system of linear equations has no solution, or has exactly one solution or has infinitely many solutions.

An arbitrary system of 'm' linear equations in 'n' unknowns can be written as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots \quad\quad \vdots \quad\quad\quad\quad \vdots \quad\quad \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

where $x_1, x_2, .., x_n$ are the unknowns and the subscripted a's and b's denote the constants.

Augmented matrices

A system of 'm' linear equations in 'n' unknowns can be abbreviated by writing only the rectangular array of numbers.

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_n \end{bmatrix}$$

This is called the augmented matrix for the system and $\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}$ is the coefficient matrix.

Consider the following system of equations

$$\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

$$\underbrace{\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 9 \\ 1 \\ 0 \end{bmatrix}}_{B}$$

$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$ is the coefficient matrix

and $[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$ is the augmented matrix.

1.1.4 Testing the consistency of non homogeneous linear equations (two and three variables) by rank method.

Consider the equations $AX = B$ in 'n' unknowns.

(i) If $\rho\left(\left[A, B\right]\right) = \rho\left(A\right)$, then the equations are consistent.

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